

EA-0015 - AQELYN Engineering Archive

AQELYN - EA-0015 Engineering Archive

IS-015 - AQELYN Security Operations (SOC) Engine

****Archive ID:**** EA-0015

****Implementation Specification:**** IS-015

****Component:**** AQELYN Security Operations (SOC) Engine

****Project:**** AQELYN

****System Type:**** Cyber Security Operating Environment

****Status:**** COMPLETE

****Repository Impact:**** No top-level repository structure changes

****Breaking Changes:**** None

****Engineering Phase:**** Phase 3

****Predecessor Archives:**** EA-0001 through EA-0014

****Next Specification:**** IS-016

Document Control

| Field | Value |

---|---

| Document | Engineering Archive EA-0015 |

| Specification | IS-015 - AQELYN Security Operations (SOC) Engine |

| Publication Format | Markdown, PDF, HTML, ZIP |

| Source of Truth | MD/EA-0015.md |

| Archive Rule | Implementation Specification -> Engineering Archive -> Continue |

| Repository Rule | Fixed repository structure; no redesign |

| Completion State | IS-015 complete; EA-0015 generated |

1. Engineering Context

AQELYN is being built as a modular Cyber Security Operating Environment.

Completed engineering archive chain:

| Engineering Archive | Implementation Specification | Component |

---|---|---

| EA-0001 | IS-001 | AQELYN Kernel |

| EA-0002 | IS-002 | Universal Object Model |

| EA-0003 | IS-003 | AQELYN Event Bus |

EA-0004	IS-004	AQELYN Evidence Engine
EA-0005	IS-005	AQELYN Knowledge Graph
EA-0006	IS-006	AQELYN Trust Engine
EA-0007	IS-007	AQELYN Mission Engine
EA-0008	IS-008	AQELYN Workflow Engine
EA-0009	IS-009	AQELYN Policy Engine
EA-0010	IS-010	AQELYN Compliance & Governance Engine
EA-0011	IS-011	AQELYN Identity & Access Governance Engine
EA-0012	IS-012	AQELYN Asset & Configuration Governance Engine
EA-0013	IS-013	AQELYN Risk Intelligence Engine
EA-0014	IS-014	AQELYN Threat Intelligence Fusion Engine
EA-0015	IS-015	AQELYN Security Operations (SOC) Engine

The fixed repository structure remains:

```
AQELYN/  
■■■■ archive/  
■■■■ blueprint/  
■■■■ docs/  
■■■■ src/  
■■■■ tests/  
■■■■ tools/  
■■■■ build/  
■■■■ releases/  
■■■■ scripts/  
■■■■ assets/  
■■■■ examples/  
■■■■ plugins/  
■■■■ sdk/  
■■■■ api/  
■■■■ README.md
```

The engineering rule remains:

```
Finish Implementation Specification  
↓  
Generate Engineering Archive  
↓  
Continue
```

No Engineering Archive may be skipped.

2. IS-015 Specification Identity

```
Specification ID: IS-015  
Name: AQELYN Security Operations (SOC) Engine  
Engineering Archive Target: EA-0015  
Project: AQELYN  
System Type: Cyber Security Operating Environment  
Status: Complete  
Predecessor: IS-014 - AQELYN Threat Intelligence Fusion Engine
```

3. Purpose

The AQELYN Security Operations (SOC) Engine coordinates security monitoring, triage, investigations, incident handling, analyst workflows, and operational response across the entire AQELYN platform.

It transforms intelligence into real-time security operations.

It answers:

What is happening now?
Which events require investigation?
Which incidents are active?
Which assets are under attack?
Which missions are affected?
Which analysts are assigned?
What evidence supports each incident?
What actions have been taken?

4. Mission

The engine shall provide:

Security monitoring
Alert management
Incident management
Case management
SOC investigations
Analyst collaboration
Threat hunting
Evidence collection
Alert prioritization
Risk-aware investigations
Mission-aware investigations
Playbook execution
Response coordination
Executive SOC dashboards

5. Scope

5.1 In Scope

SOC dashboards
Alert ingestion
Alert normalization
Alert correlation
Security incidents
Incident lifecycle
Case management
Investigation timelines
Analyst assignments
Evidence linking
Threat hunting
SOC reporting
Workflow integration
Mission impact awareness

5.2 Out of Scope

Endpoint antivirus engine
Network IDS implementation
Firewall implementation
Email gateway implementation
SIEM storage engine
Physical security operations
Call-center ticketing

6. Dependencies

IS-015 depends on:

IS-001 AQELYN Kernel
IS-002 Universal Object Model
IS-003 AQELYN Event Bus

```
IS-004 AQELYN Evidence Engine
IS-005 AQELYN Knowledge Graph
IS-006 AQELYN Trust Engine
IS-007 AQELYN Mission Engine
IS-008 AQELYN Workflow Engine
IS-009 AQELYN Policy Engine
IS-010 AQELYN Compliance & Governance Engine
IS-011 Identity & Access Governance Engine
IS-012 Asset & Configuration Governance Engine
IS-013 Risk Intelligence Engine
IS-014 Threat Intelligence Fusion Engine
```

7. High-Level Architecture

```
AQELYN Security Operations Engine
■
■■■ Alert Manager
■■■ Incident Manager
■■■ Case Manager
■■■ Investigation Engine
■■■ Threat Hunting Engine
■■■ Analyst Workspace
■■■ Playbook Engine
■■■ Response Coordinator
■■■ SOC Reporting Service
■■■ Mission Impact Connector
■■■ Risk Connector
■■■ Evidence Binder
■■■ Event Publisher
```

8. Functional Requirements

FR-015-001 - Alert Manager

The engine shall receive alerts from internal AQELYN engines and approved external integrations.

Each alert shall include:

```
alert_id
source
severity
confidence
status
timestamp
related_assets
related_identities
related_missions
evidence_links
```

FR-015-002 - Incident Manager

The engine shall convert correlated alerts into managed security incidents.

Each incident shall include:

```
incident_id
title
priority
severity
status
owner
affected_assets
affected_missions
risk_score
threat_context
evidence
```

FR-015-003 - Case Management

Each investigation shall be maintained as a governed case.

Supported capabilities:

- Case creation
- Task assignment
- Investigation timeline
- Evidence collection
- Notes
- Attachments
- Workflow integration
- Closure review

FR-015-004 - Investigation Engine

The system shall support structured investigations using evidence from:

- Event Bus
- Evidence Engine
- Knowledge Graph
- Threat Intelligence
- Risk Intelligence
- Asset Governance
- Identity Governance
- Mission Engine

FR-015-005 - Threat Hunting

The engine shall support proactive threat hunting using:

- Indicators
- Threat actors
- Campaigns
- TTPs
- Behavioral analytics
- Mission context
- Risk intelligence

FR-015-006 - Analyst Workspace

Provide a unified operational workspace for SOC analysts including:

- Alert queue
- Incident queue
- Case management
- Evidence explorer
- Knowledge graph navigation
- Threat intelligence view
- Risk dashboard
- Mission impact view

FR-015-007 - Playbook Execution

The engine shall execute standardized response playbooks for:

- Malware
- Phishing
- Credential compromise
- Insider threat
- Data exfiltration
- Ransomware
- Supply-chain attack
- Privilege escalation

FR-015-008 - Response Coordination

Coordinate:

Analysts
Approvers
Mission owners
Asset owners
Risk owners
Compliance officers

FR-015-009 - SOC Reporting

Generate reports including:

Open incidents
Incident trends
Response times
Mean Time To Detect (MTTD)
Mean Time To Respond (MTTR)
Mission impact
Risk reduction
Threat activity
Playbook effectiveness

FR-015-010 - Event Publication

Publish standardized operational events:

alert.created
alert.updated
incident.created
incident.closed
case.created
case.closed
hunt.completed
playbook.executed

9. Non-Functional Requirements

The engine shall provide:

Continuous monitoring
Evidence-backed investigations
Real-time alert handling
Auditability
Analyst traceability
Workflow integration
High availability
Scalability
Repository stability
Backward compatibility

10. Core SOC Workflow

Alert received
↓
Alert normalized
↓
Correlation
↓
Incident created
↓
Case opened
↓
Evidence collected
↓
Threat hunting
↓

```
Response playbook executed
↓
Mission/Risk updated
↓
Incident closed
```

11. Internal Component Architecture

The AQELYN Security Operations Engine is implemented as a modular, event-driven operational subsystem built on the AQELYN Kernel and Event Bus.

```
AQELYN Security Operations Engine
■
■■■ Alert Manager
■■■ Alert Correlation Engine
■■■ Incident Manager
■■■ Case Manager
■■■ Investigation Engine
■■■ Threat Hunting Engine
■■■ Analyst Workspace
■■■ Playbook Engine
■■■ Response Coordinator
■■■ Evidence Binder
■■■ Mission Impact Connector
■■■ Risk Connector
■■■ Executive Dashboard
■■■ Event Publisher
```

Each module operates independently while sharing the Universal Object Model, Evidence Engine, Knowledge Graph, and Event Bus.

12. Component Specifications

12.1 Alert Manager

Responsible for receiving, validating, deduplicating, and classifying alerts.

Responsibilities:

```
Alert ingestion
Alert validation
Alert normalization
Alert prioritization
Alert deduplication
Alert enrichment
Alert lifecycle
```

12.2 Alert Correlation Engine

Correlates alerts originating from multiple AQELYN engines.

Correlation factors:

```
Shared asset
Shared identity
Shared mission
Shared threat
Shared evidence
Time proximity
Behavior similarity
```

12.3 Incident Manager

Creates governed incidents from correlated alerts.

Responsibilities:

- Incident creation
- Incident prioritization
- Severity calculation
- Assignment
- Lifecycle tracking
- Closure validation

12.4 Case Manager

Maintains investigation cases.

Capabilities:

- Case creation
- Task management
- Evidence attachment
- Timeline
- Notes
- Review workflow
- Closure approval

12.5 Investigation Engine

Provides structured investigation workflows.

Investigation sources:

- Evidence Engine
- Knowledge Graph
- Threat Intelligence
- Risk Intelligence
- Identity Governance
- Asset Governance
- Mission Engine
- Workflow Engine

12.6 Threat Hunting Engine

Supports proactive hunting activities.

Hunting methods:

- Indicator hunting
- Behavior hunting
- Threat actor hunting
- Campaign hunting
- Mission hunting
- Risk hunting

12.7 Analyst Workspace

Unified operational interface.

Views include:

- Alert Queue
- Incident Queue
- Case Queue
- Evidence Explorer
- Threat Intelligence
- Risk Dashboard
- Mission Dashboard

Knowledge Graph

12.8 Playbook Engine

Executes standardized response procedures.

Supported playbooks:

- Malware Response
- Phishing Response
- Credential Theft
- Privilege Escalation
- Insider Threat
- Ransomware
- Data Exfiltration
- Supply Chain Attack
- Cloud Compromise

12.9 Response Coordinator

Coordinates human and automated response activities.

Coordinates:

- SOC Analysts
- Incident Commanders
- Asset Owners
- Mission Owners
- Risk Owners
- Compliance Officers
- Automation Services

12.10 Evidence Binder

Associates every operational action with immutable evidence.

Evidence sources:

- Alerts
- Incidents
- Cases
- Threat Intelligence
- Risk Assessments
- Asset Records
- Identity Records
- Mission Records

12.11 Executive Dashboard

Provides executive operational awareness.

Displays:

- Current incidents
- Critical alerts
- Mission impact
- Enterprise risk
- Threat landscape
- SOC workload
- Response metrics

13. Universal Object Model Extensions

13.1 Alert

```
Alert:
alert_id
source
severity
confidence
timestamp
evidence
```

13.2 Incident

```
Incident:
incident_id
priority
severity
owner
status
evidence
```

13.3 Case

```
Case:
case_id
incident
owner
tasks
evidence
```

13.4 Investigation

```
Investigation:
investigation_id
case_id
findings
evidence
```

13.5 PlaybookExecution

```
PlaybookExecution:
execution_id
playbook
incident
status
```

14. Knowledge Graph Integration

Relationships include:

```
Alert
↓
creates
↓
Incident
Incident
↓
contains
↓
Case
Case
↓
contains
↓
Investigation
Incident
↓
affects
```

```
↓  
Mission  
Incident  
↓  
increases  
↓  
Risk  
Incident  
↓  
supported_by  
↓  
Evidence  
Threat  
↓  
causes  
↓  
Incident
```

15. Event Bus Integration

15.1 Alert Events

```
alert.created  
alert.updated  
alert.closed
```

15.2 Incident Events

```
incident.created  
incident.updated  
incident.closed  
incident.escalated
```

15.3 Case Events

```
case.created  
case.updated  
case.closed
```

15.4 Investigation Events

```
investigation.started  
investigation.updated  
investigation.completed
```

15.5 Playbook Events

```
playbook.started  
playbook.completed  
playbook.failed
```

15.6 SOC Events

```
soc.dashboard.updated  
soc.metrics.updated
```

16. Evidence Engine Integration

The SOC Engine consumes evidence from:

- Alerts
- Threat Intelligence
- Risk Intelligence
- Asset Governance
- Identity Governance
- Mission Engine
- Compliance Engine
- Workflow Engine

Evidence shall remain immutable after publication.

17. Policy Engine Integration

Policies govern:

- Incident severity
- Escalation rules
- Playbook authorization
- Closure approval
- Assignment rules
- Retention periods

18. Threat Intelligence Integration

Consumes:

- Indicators
- Threat Actors
- Campaigns
- Confidence Scores
- Threat Reports

Threat Intelligence provides investigation context.

19. Risk Intelligence Integration

Consumes:

- Risk Scores
- Risk Trends
- Mission Risk
- Risk Treatments

SOC investigations dynamically update enterprise risk.

20. Mission Engine Integration

Provides:

- Mission priority
- Mission dependency
- Mission impact
- Operational criticality

Mission context influences incident prioritization.

21. Compliance Integration

SOC investigations reference:

- Compliance findings
- Control failures
- Exceptions
- Assessment history

22. Asset Governance Integration

Provides:

- Asset inventory
- Configuration drift
- Criticality
- Ownership
- Exposure

23. Identity Governance Integration

Provides:

- Privileged identities
- Access anomalies
- Ownership
- Credential exposure

24. Public APIs

24.1 Alert API

- GET /alerts
- POST /alerts
- GET /alerts/{id}
- PUT /alerts/{id}

24.2 Incident API

- GET /incidents
- POST /incidents
- GET /incidents/{id}
- PUT /incidents/{id}

24.3 Case API

- GET /cases
- POST /cases
- GET /cases/{id}

24.4 Investigation API

- GET /investigations
- POST /investigations

24.5 Playbook API

```
GET /playbooks
POST /playbooks/execute
```

24.6 Dashboard API

```
GET /soc/dashboard
GET /soc/metrics
```

25. Repository Impact

Implementation shall follow the approved AQELYN repository.

```
AQELYN/
■■■ src/
■ ■■■ security_operations/
■■■ tests/
■ ■■■ security_operations/
■■■ docs/
■ ■■■ security_operations/
■■■ api/
■ ■■■ security_operations/
■■■ archive/
```

No top-level repository modifications are permitted.

26. Security Architecture

The SOC Engine is a Tier-1 operational component responsible for coordinating all security investigations and incident response activities.

Every operational decision shall be:

```
Evidence-backed
Policy-governed
Auditable
Traceable
Explainable
Role-authorized
Mission-aware
Risk-aware
```

26.1 Security Principles

The engine shall implement:

```
Zero Trust
Least Privilege
Separation of Duties
Immutable Evidence
Policy Enforcement
Continuous Monitoring
Defense in Depth
Secure by Design
```

26.2 Authorization Model

Supported operational roles:

```
SOC Analyst
Senior Analyst
Incident Commander
Threat Hunter
Mission Owner
```

```
Asset Owner
Risk Owner
Compliance Officer
Security Administrator
Automation Service
```

All privileged actions shall be validated through the AQELYN Policy Engine.

26.3 Operational Integrity

Operational objects shall maintain:

```
Unique identifier
Version history
Evidence references
Actor attribution
Policy decision references
Timeline history
Closure rationale
Audit state
```

Incident and case history shall be append-only.

26.4 Evidence Protection

SOC evidence shall support:

```
Immutable storage
Evidence lineage
Chain-of-custody references
Access logging
Integrity verification
Trust scoring
Audit history
```

The SOC Engine shall not overwrite source evidence.

27. SOC Operational Lifecycle

27.1 Alert Lifecycle

```
Received
↓
Validated
↓
Normalized
↓
Correlated
↓
Prioritized
↓
Closed
```

27.2 Incident Lifecycle

```
Created
↓
Assigned
↓
Investigated
↓
Contained
↓
Recovered
↓
Closed
```

↓
Archived

27.3 Case Lifecycle

Opened
↓
Evidence Collected
↓
Analysis
↓
Review
↓
Approved
↓
Closed

27.4 Investigation Lifecycle

Started
↓
Evidence Collection
↓
Correlation
↓
Threat Analysis
↓
Findings
↓
Recommendations
↓
Completed

27.5 Playbook Lifecycle

Selected
↓
Authorized
↓
Executed
↓
Verified
↓
Completed

28. Continuous Monitoring

The engine continuously evaluates:

New alerts
Incident changes
Evidence updates
Threat intelligence updates
Risk changes
Mission impact
Identity exposure
Asset exposure
Compliance findings
Workflow completion

Significant changes may trigger alert enrichment, incident escalation, case updates, playbook execution, or risk/mission updates.

29. Performance Requirements

The SOC Engine shall support:

- Real-time alert processing
- High-volume incident creation
- Concurrent investigations
- Parallel playbook execution
- Scalable dashboards
- Low-latency event publication

SOC processing shall not block the AQELYN Event Bus or upstream engines.

30. Scalability Requirements

The engine shall scale to support:

- Millions of alerts
- Hundreds of thousands of incidents
- Large case repositories
- Distributed analyst teams
- Multi-tenant environments
- Hybrid deployments

31. Audit Requirements

Every operational action shall generate immutable audit records.

Audit events include:

- Alert creation
- Incident assignment
- Case updates
- Evidence access
- Playbook execution
- Policy decisions
- Investigation completion

32. Failure Handling

32.1 Alert Processing Failure

- Alert queued
- Retry performed
- Audit recorded

32.2 Incident Creation Failure

- Incident pending
- Evidence retained
- Retry scheduled

32.3 Event Bus Failure

- Events persisted
- Delivery retried
- Ordering preserved

32.4 Evidence Failure

- Investigation suspended
- Analyst notified

Policy enforced

33. Testing Strategy

33.1 Unit Testing

Validate:

Alert Manager
Incident Manager
Case Manager
Investigation Engine
Threat Hunting Engine
Playbook Engine
Response Coordinator
Dashboard Services

33.2 Integration Testing

Verify interaction with:

Kernel
Event Bus
Evidence Engine
Knowledge Graph
Trust Engine
Mission Engine
Workflow Engine
Policy Engine
Compliance Engine
Identity Governance
Asset Governance
Risk Intelligence
Threat Intelligence

33.3 System Testing

Validate:

Alert processing
Incident management
Case lifecycle
Threat hunting
Playbook execution
Mission updates
Risk updates
Executive dashboards

33.4 Security Testing

Verify:

Authorization
Policy enforcement
Audit completeness
Evidence protection
Workflow security

33.5 Regression Testing

Verify IS-001 through IS-014 remain unaffected.

34. Acceptance Criteria

IS-015 is complete when:

```
Alert Manager implemented
Incident Manager implemented
Case Manager implemented
Investigation Engine implemented
Threat Hunting Engine implemented
Playbook Engine implemented
Response Coordinator implemented
Executive Dashboard defined
Integrations completed
Repository unchanged
Testing documented
```

35. Repository Validation

Repository structure remains fixed.

```
AQELYN/
■■■ src/security_operations/
■■■ tests/security_operations/
■■■ docs/security_operations/
■■■ api/security_operations/
■■■ archive/
```

No top-level changes are permitted.

36. Engineering Summary

IS-015 introduces the operational Security Operations Center capability for AQELYN.

Major capabilities:

```
Alert Management
Incident Management
Case Management
Threat Hunting
SOC Investigations
Playbook Execution
Response Coordination
Executive Dashboards
Mission-aware Operations
Risk-aware Operations
Evidence-backed Investigations
```

The design preserves modularity, repository stability, event-driven integration, and backward compatibility.

37. Specification Status

```
Specification ID : IS-015
Title : AQELYN Security Operations (SOC) Engine
Status : COMPLETE
Engineering Archive : READY FOR GENERATION
Next Artifact : EA-0015
```

Engineering workflow status:

```
EA-0001 COMPLETE
EA-0002 COMPLETE
```

```
EA-0003 COMPLETE
EA-0004 COMPLETE
EA-0005 COMPLETE
EA-0006 COMPLETE
EA-0007 COMPLETE
EA-0008 COMPLETE
EA-0009 COMPLETE
EA-0010 COMPLETE
EA-0011 COMPLETE
EA-0012 COMPLETE
EA-0013 COMPLETE
EA-0014 COMPLETE
IS-015 COMPLETE
EA-0015 READY FOR GENERATION
```

38. EA-0015 Engineering Objective

The objective of IS-015 was to introduce a dedicated Security Operations Engine that enables AQELYN to coordinate monitoring, alert triage, incident management, case management, investigations, threat hunting, response playbooks, analyst workflows, and operational reporting.

The engine extends AQELYN from intelligence and governance into real-time operational security response.

39. EA-0015 Engineering Summary

The implementation specification defines a modular subsystem responsible for:

```
Alert management
Alert correlation
Incident management
Case management
Investigation workflows
Threat hunting
Analyst workspace
Playbook execution
Response coordination
Evidence binding
Mission impact awareness
Risk intelligence integration
SOC reporting
Event publishing
```

The engine integrates with all previously completed AQELYN engines while preserving architectural modularity.

40. Major Engineering Decisions

40.1 Decision 1 - Dedicated SOC Engine

SOC operational responsibilities are implemented as a standalone engine rather than embedded in Risk Intelligence, Threat Intelligence, or Workflow.

Rationale:

```
Clear separation of operations from analytics.
Independent lifecycle and scaling.
Better support for analyst workflows and incident response.
Improved operational accountability and auditability.
```

40.2 Decision 2 - Evidence-Backed Investigations

Every alert, incident, case, investigation, and playbook execution shall reference immutable evidence.

Benefits:

SOC decisions become auditable.
Investigations are traceable.
Incident closure can be verified.
Risk and mission updates can be supported by proof.

40.3 Decision 3 - Mission-Aware and Risk-Aware Operations

Incident prioritization is informed by mission and risk context.

Benefits:

Operational attention aligns with mission impact.
Enterprise risk influences response urgency.
High-impact events can be escalated automatically.
Executive dashboards reflect operational consequences.

40.4 Decision 4 - Event-Driven SOC Operations

Alert, incident, case, investigation, playbook, dashboard, and metric updates are published through the AQELYN Event Bus.

Examples include:

alert.created
alert.updated
incident.created
incident.closed
case.created
case.closed
investigation.started
investigation.completed
playbook.started
playbook.completed
soc.metrics.updated

This maintains loose coupling between AQELYN engines.

40.5 Decision 5 - Universal Object Model Extension

New domain objects introduced include:

Alert
Incident
Case
Investigation
PlaybookExecution

These extend the Universal Object Model without modifying existing object definitions.

41. Architectural Integration Summary

| Engine | Integration |

|---|---|

| IS-001 Kernel | Runtime lifecycle and service registration |

| IS-002 Universal Object Model | Alert, incident, case, investigation, playbook objects |

| IS-003 Event Bus | Alert, incident, case, investigation, playbook, SOC events |

IS-004 Evidence Engine	Immutable SOC evidence, investigation evidence, case evidence
IS-005 Knowledge Graph	Alert, incident, case, threat, risk, mission, evidence relationships
IS-006 Trust Engine	Evidence confidence and operational confidence
IS-007 Mission Engine	Mission priority, dependency, impact, criticality
IS-008 Workflow Engine	Escalation, case review, response, playbook workflows
IS-009 Policy Engine	Severity, escalation, closure, assignment, retention, authorization
IS-010 Compliance Engine	Compliance findings, control failures, exceptions, assessment history
IS-011 Identity Governance Engine	Privileged identities, access anomalies, credential exposure
IS-012 Asset Governance Engine	Asset inventory, drift, criticality, ownership, exposure
IS-013 Risk Intelligence Engine	Risk score, trend, treatment, mission risk
IS-014 Threat Intelligence Engine	Indicators, actors, campaigns, confidence, reports

No existing engine required redesign.

42. Repository Impact Summary

Repository structure remains unchanged.

Implementation is expected within existing project directories, including:

```
AQELYN/  
■■■ src/security_operations/  
■■■ tests/security_operations/  
■■■ api/security_operations/  
■■■ docs/security_operations/  
■■■ archive/
```

No top-level directories were added, removed, or renamed.

43. Security Impact Summary

The specification introduces SOC-specific security controls:

```
Policy-driven operational authorization  
Role-based analyst operations  
Evidence-backed investigations  
Immutable case and incident history  
Mission-aware escalation  
Risk-aware prioritization  
Auditable playbook execution  
Separation of duties for incident closure  
Traceable analyst actions
```

No reduction in the security posture of existing components was identified.

44. Capabilities Added

The engine enables AQELYN to support:

```
Alert ingestion  
Alert normalization  
Alert correlation  
Incident lifecycle management  
Case management  
Investigation timelines  
Analyst assignments
```

Evidence linking
Threat hunting
SOC reporting
Playbook execution
Response coordination
Mission impact awareness
Risk-aware operations
Executive SOC dashboards

45. Risks Identified

| Risk | Mitigation |

|---|---|

| Alert overload | Alert correlation, prioritization, deduplication |

| Incident misclassification | Policy-based severity and evidence-backed assignment |

| Evidence gaps | Evidence Binder and workflow-driven evidence collection |

| Unauthorized playbook execution | Policy enforcement and role authorization |

| Analyst action ambiguity | Immutable audit records and actor attribution |

| Mission impact underestimation | Mission Engine integration |

| Risk context drift | Risk Intelligence integration and continuous monitoring |

| Case closure without proof | Closure validation and evidence requirements |

No critical architectural risks were identified that require redesign.

46. Verification Summary

The specification defines verification for:

Unit testing
Integration testing
System testing
Security testing
Regression testing

Acceptance criteria cover alert management, incident management, case management, investigation engine, threat hunting, playbook execution, response coordination, executive dashboards, integrations, repository validation, and testing documentation.

47. Engineering Principles Confirmed

The implementation complies with established AQELYN principles:

Modular architecture
Event-driven communication
Immutable evidence references
Traceability
Explainability
Security by design
Repository stability
Backward compatibility
Governance-before-archive discipline

48. Dependencies

Required:

EA-0001 through EA-0014
IS-001 through IS-014

Enables:

IS-016 and subsequent operations-dependent components

49. Completion Record

Engineering Archive : EA-0015
Implementation Specification : IS-015
Title : AQELYN Security Operations (SOC) Engine
Engineering Status : COMPLETE
Repository Status : UNCHANGED
Architecture Status : EXTENDED
Backward Compatibility : MAINTAINED
Engineering Rule :
IS Completed
↓
EA Generated
↓
Continue

50. Archive Index Update

EA-0001 IS-001 AQELYN Kernel
EA-0002 IS-002 Universal Object Model
EA-0003 IS-003 AQELYN Event Bus
EA-0004 IS-004 AQELYN Evidence Engine
EA-0005 IS-005 AQELYN Knowledge Graph
EA-0006 IS-006 AQELYN Trust Engine
EA-0007 IS-007 AQELYN Mission Engine
EA-0008 IS-008 AQELYN Workflow Engine
EA-0009 IS-009 AQELYN Policy Engine
EA-0010 IS-010 AQELYN Compliance & Governance Engine
EA-0011 IS-011 AQELYN Identity & Access Governance Engine
EA-0012 IS-012 AQELYN Asset & Configuration Governance Engine
EA-0013 IS-013 AQELYN Risk Intelligence Engine
EA-0014 IS-014 AQELYN Threat Intelligence Fusion Engine
EA-0015 IS-015 AQELYN Security Operations (SOC) Engine

51. Engineering Phase Status

Completed Engineering Archives : EA-0001 through EA-0015

Current Status:
EA-0015 COMPLETE

Next Implementation Specification:
IS-016

EA-0015 is completed and archived. The engineering workflow is consistent with the project rule:

Implementation Specification -> Engineering Archive -> Continue

From this point onward, the next engineering artifact is IS-016.

52. Engineering Archive Publication Standard

EA-0015 follows the AQELYN Engineering Archive Publication Standard.

Required package structure:

```
EA-xxxx/  
■  
■ ■ ■ ■ diagrams/  
■ ■ ■ ■ Architecture.svg  
■ ■ ■ ■ Component.svg  
■ ■ ■ ■ Workflow.svg  
■ ■ ■ ■ EventFlow.svg  
■ ■ ■ ■ Integration.svg  
■  
■ ■ ■ ■ examples/  
■ ■ ■ ■ example_*.md  
■  
■ ■ ■ ■ HTML/  
■ ■ ■ ■ index.html  
■ ■ ■ ■ styles.css  
■ ■ ■ ■ assets/  
■  
■ ■ ■ ■ images/  
■  
■ ■ ■ ■ journal/  
■ ■ ■ ■ Engineering_Journal.md  
■  
■ ■ ■ ■ manifest/  
■ ■ ■ ■ manifest.json  
■  
■ ■ ■ ■ MD/  
■ ■ ■ ■ EA-xxxx.md  
■  
■ ■ ■ ■ PDF/  
■ ■ ■ ■ EA-xxxx.pdf  
■  
■ ■ ■ ■ requirements/  
■ ■ ■ ■ Requirements_Matrix.md  
■  
■ ■ ■ ■ traceability/  
■ ■ ■ ■ Traceability_Matrix.md  
■  
■ ■ ■ ■ README.md
```

The master Markdown is the source of truth. The PDF and HTML are generated from the same master Markdown and must not omit sections.

53. Requirements Matrix

Requirement ID Requirement Evidence in Archive Status			
--- --- --- ---			
FR-015-001	Receive and manage alerts	Sections 8, 12	Complete
FR-015-002	Convert correlated alerts to incidents	Sections 8, 12	Complete
FR-015-003	Maintain governed cases	Sections 8, 12, 27	Complete
FR-015-004	Support structured investigations	Sections 8, 12, 27	Complete
FR-015-005	Support threat hunting	Sections 8, 12	Complete
FR-015-006	Provide analyst workspace	Sections 8, 12	Complete
FR-015-007	Execute response playbooks	Sections 8, 12, 27	Complete
FR-015-008	Coordinate response roles	Sections 8, 12	Complete
FR-015-009	Generate SOC reports	Sections 8, 12, 31	Complete
FR-015-010	Publish SOC events	Sections 8, 15, 40	Complete
NFR-015-001	Continuous monitoring	Sections 9, 28	Complete
NFR-015-002	Evidence-backed investigations	Sections 9, 16, 40	Complete
NFR-015-003	Real-time alert handling	Sections 9, 29	Complete

| NFR-015-004 | Auditability | Sections 9, 31, 43 | Complete |

| NFR-015-005 | Workflow integration | Sections 9, 17, 41 | Complete |

| NFR-015-006 | Repository stability | Sections 25, 35, 42 | Complete |

54. Traceability Matrix

| Source | Target | Relationship |

---|---|---

| IS-015 Purpose | EA-0015 Objective | Defines why the engine exists |

| Alert Manager | FR-015-001 | Implements alert lifecycle |

| Alert Correlation Engine | FR-015-002 | Supports incident creation |

| Incident Manager | FR-015-002 | Implements incident lifecycle |

| Case Manager | FR-015-003 | Implements case management |

| Investigation Engine | FR-015-004 | Implements structured investigations |

| Threat Hunting Engine | FR-015-005 | Implements proactive hunting |

| Analyst Workspace | FR-015-006 | Implements analyst operational view |

| Playbook Engine | FR-015-007 | Implements response playbooks |

| Response Coordinator | FR-015-008 | Coordinates operational response |

| Executive Dashboard | FR-015-009 | Implements SOC reporting |

| Event Publisher | FR-015-010 | Publishes SOC events |

| Evidence Engine Integration | Evidence-backed investigations | References immutable evidence |

| Policy Engine Integration | Severity and authorization rules | Determines escalation and closure behavior |

| Mission Engine Integration | Mission-aware prioritization | Supplies mission impact context |

| Risk Intelligence Integration | Risk-aware investigations | Supplies risk context |

| Threat Intelligence Integration | Threat context | Supplies indicators, actors, campaigns |

| Event Bus Integration | Event-driven synchronization | Publishes operational events |

| Repository Validation | Repository Standard | Confirms no top-level redesign |

55. Engineering Journal

Journal Entry - EA-0015

EA-0015 was created to archive completion of IS-015 - AQELYN Security Operations (SOC) Engine.

The archive records the expansion of AQELYN into security operations. IS-015 defines the structure needed to manage alerts, incidents, cases, investigations, threat hunting, playbook execution, response coordination, analyst workspaces, operational reporting, and evidence-backed SOC activity.

The engineering design preserves the fixed AQELYN repository structure and maintains backward compatibility with previously completed engines.

Lessons Learned

Security operations must be modeled separately from threat intelligence and risk intelligence. Threat intelligence supplies context, risk intelligence supplies prioritization, but the SOC Engine owns operational triage, investigation, incident handling, case management, and response coordination.

Governance Note

EA-0015 follows the master-document publication workflow. The Markdown file is the authoritative source, and PDF/HTML representations are generated from the same content.

56. Examples

56.1 Example Alert

```
alert_id: ALERT-0001
source: threat_intelligence
severity: high
confidence: 0.91
timestamp: 2026-07-07T12:00:00Z
related_assets:
- ASSET-0002
related_identities:
- ID-0001
related_missions:
- MISSION-001
evidence:
- evidence://indicator-match-1001
```

56.2 Example Incident

```
incident_id: INC-1001
title: Mission-critical asset matched high-confidence threat indicator
priority: critical
severity: high
status: active
owner: soc_analyst_01
affected_assets:
- ASSET-0002
affected_missions:
- MISSION-001
risk_score: 92
threat_context:
campaign: CAMP-2001
evidence:
- evidence://indicator-match-1001
- evidence://asset-criticality-asset-0002
```

56.3 Example Case

```
case_id: CASE-2001
incident: INC-1001
owner: senior_analyst_01
tasks:
- collect_endpoint_evidence
- validate_indicator_match
- notify_mission_owner
evidence:
- evidence://case-timeline-2001
```

56.4 Example SOC Event

```
{
  "event_type": "incident.created",
  "incident_id": "INC-1001",
  "severity": "high",
  "priority": "critical",
  "reason": "High-confidence threat indicator matched mission-critical asset",
  "source_engine": "aqelyn_security_operations_engine"
}
```

57. Manifest Summary

Archive contents include:

```
README.md
MD/EA-0015.md
PDF/EA-0015.pdf
HTML/index.html
HTML/styles.css
manifest/manifest.json
requirements/Requirements_Matrix.md
traceability/Traceability_Matrix.md
journal/Engineering_Journal.md
diagrams/Architecture.svg
diagrams/Component.svg
diagrams/Workflow.svg
diagrams/EventFlow.svg
diagrams/Integration.svg
examples/example_soc.md
```

58. Final Archive Statement

EA-0015 is the Engineering Archive for IS-015 - AQELYN Security Operations (SOC) Engine.

It records the completed specification, the architectural decisions, the integration model, the repository impact, the risk posture, verification requirements, acceptance criteria, archive index update, and the engineering publication standard.

```
EA-0015 COMPLETE
IS-015 COMPLETE
NEXT: IS-016
```